INCH-POUND
MIL-F-24724(SH)
15 February 1990

MILITARY SPECIFICATION

FILTER HOUSINGS (HYDRAULIC) AND DIFFERENTIAL PRESSURE INDICATORS, GENERAL SPECIFICATION FOR

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers hydraulic filter housings with differential pressure actuated bypass valves and connected indicators for use in ship hydraulic systems with an operating pressure up to 3,000 pounds per square inch (lb/in^2) as specified in the applicable specification sheets (see 3.1).

1.2 Classification. Filter housings shall be of the envelope sizes specified in applicable specification sheets (see 3.1). The standard part numbers for the filter housings shall be as specified in the applicable specification sheets (see 3.1).

2. APPLICABLE DOCUMENTS

2.1 Government document

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specification and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4330
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:20Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

PPP-F-320	- Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes.

MILITARY

MIL-S-901	- Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for.
MIL-C-5501	- Caps and Plugs, Protective, Dust and Moisture Seal, General Specification for.
MIL-H-6083	- Hydraulic Fluid, Petroleum Base, for Preservation and Operation.
MIL-H-17672	- Hydraulic Fluid, Petroleum, Inhibited.
MIL-L-19140	- Lumber and Plywood, Fire-Retardant Treated.
MIL-F-24702	- Filter Elements, Hydraulic, Disposable, General Specification for (Metric)
MIL-F-24702/1	- Filter Elements, Hydraulic, Disposable.
MIL-F-24724/1	- Filter Housings (Hydraulic) and Differential Pressure Indicators (1, 2, 3 Elements).
MIL-F-24724/2	- Filter Housings (Hydraulic) and Differential Pressure Indicators (4 and 6 Elements).

STANDARDS

MILITARY

MIL-STD-278	- Welding and Casting Standard.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.
MIL-STD-2073-1	- DOD Materiel Procedures for Development and Application of Packaging Requirements.
MIL-STD-2193	- Hydraulic System Components, Ship. (Metric)

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:20Z
Check the source to verify that this is the current version before use.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
Y14.5 - Dimensioning and Tolerancing. (DOD adopted)
Y32.10 - Graphic Symbols for Fluid Power Diagrams. (DOD adopted)

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
D 3951 - Standard Practice for Commercial Packaging. (DOD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL FLUID POWER ASSOCIATION (NFP(A))
T2.6.1 - Method for Verifying the Fatigue and Static Pressure Ratings of the Pressure Containing Envelope of a Metal Fluid Power Component.

(Application for copies should be addressed to the National Fluid Power Association, 3333 N. Mayfair Road, Milwaukee, WI 53222.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)
J 1926 - Specification for Straight Thread O-Ring Boss Port.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

(Non-Government standards and other publications are normally available from the organizations that prepare or that distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:20Z
Check the source to verify that this is the current version before use.

3.2 Qualification. The filter housings furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time of award of contract (see 4.3 and 6.3).

3.3 Materials. Materials shall be in accordance with MIL-STD-2193. Where threaded connections are used between the filter bowl and housing, the thread design and materials shall be selected to prevent galling. The bowl and head shall be either constructed of dissimilar materials which do not gall when in contact, or of similar materials which have a hardness difference of at least 45 Brinell points. Plating or coating threads to achieve hardness difference shall not be permitted.

3.3.1 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.3.2 Compatibility. The filter housing components shall be compatible with the fluids identified in the compatibility requirements specified in MIL-STD-2193.

3.4 Design and construction. Unless otherwise specified herein, the design, parts and construction of filter housings shall be in accordance with the construction requirements specified herein, MIL-STD-2193, and in the applicable specification sheets (see 3.1).

3.4.1 Filter housings. Filter housings shall be of the full-flow type. With the filter element installed, the oil flow through the housing shall be "outside-in". The fluid entering the housing shall not impinge directly upon the element filter medium. The housings shall be of a construction that allows the filter elements to be removed for service and examination without disconnecting fittings or disturbing the mounting. The construction shall be such that the filter elements cannot be improperly installed. The filter housings shall be designed for the flowrates specified in the applicable specification sheets (see 3.1). These flowrates are based on the use of fluid in accordance with MIL-L-17672. When fluids of lower viscosities are used, the filter assemblies will be suitable for higher flowrates.

3.4.2 Structural strength. Filter housings shall withstand all the structural loads imposed by the functional test requirements of this specification. In addition, the filter housing and mountings shall be of such strength and rigidity as to withstand the torque loads required for making tube and pipe connections and replacing filter elements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:20Z
Check the source to verify that this is the current version before use.

3.4.3 Porting block. Integral to the porting block of the filter assembly shall be a differential pressure indicator and bypass valve. The indicator and bypass valve shall permit repair or replacement without replacing the porting block or housing.

3.4.4 Seals. Seals and seal glands shall be in accordance with MIL-STD-2193.

3.4.5 Fasteners. Screws, bolts and threaded inserts shall be in accordance with MIL-STD-2193.

3.4.6 Dimensions. The filter housing dimensions shall be as specified (see 3.1).

3.4.7 Protective devices.

3.4.7.1 Bypass-relief valve. The filter housings shall incorporate a relief valve to automatically bypass hydraulic fluid in the event of excessive flow restriction through the filter elements. The cracking, reseat, and full flow differential pressure across the relief valve shall conform to the values specified (see 3.1 and 6.2).

3.4.7.2 Differential pressure indicators. Each filter housing porting block shall incorporate an integral differential pressure indicator (see 3.4.3 and 6.2). The indicator shall show, by mechanical or electrical means (see 3.4.7.2.1 and 3.4.7.2.2), the position of the filter bypass valve. The indicator shall not inadvertently activate under vibration of a 20 gravity shock load. In the absence of flow through the filter, the indicators shall reset automatically and shall be manually checked for visual pointer rotation over all settings on a tricolor scale (see 3.4.7.2.1).

3.4.7.2.1 Mechanical indicators. Mechanical pressure indicators (style M) shall actuate, by raising or moving an indicator, at a filter element differential pressure value specified (see 3.1). Mechanical pressure indicators shall provide a visual indication of the bypass valve position concurrent with the filter element condition on a tricolor scale. A green area of the scale shall represent the "closed" position of the bypass and "clean" condition of the filter element; a yellow area shall represent the "closed" position of the bypass valve and "becoming contaminated" condition of the filter element; a red area of the scale shall represent the "cracked to fully open" position of the bypass valve and "clogged" condition of the filter element. The junction of green and yellow area and corresponding indicator pointer position shall represent 80 percent of the cracking pressure setting. The junction of yellow and red area and corresponding indicator pointer position shall represent 98 percent of the minimum cracking pressure setting specified (see 3.1). Unless otherwise specified (see 3.1), these settings shall have a tolerance of plus or minus 3 lb/in^2.

3.4.7.2.2 Electrical indicators. Electrical pressure indicators (style E) shall raise or move an indicator, at a filter element differential pressure value specified (see 3.1). The indicator shall provide an electrical and visual indication of the bypass valve position concurrent with the filter element condition. The visual indication shall be on a tricolor scale as described in

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:20Z
Check the source to verify that this is the current version before use.

3.4.7.2.1. The electrical indication shall be either alarm or light signal. In an operating system, the signal shall be continuously activated when the bypass valve is in the open position. The electric switch shall be enclosed with terminals in the switch housing. Electrical rating on a single-pole, double-throw switch shall be 5 amperes at 110 volts alternating current (Vac), 28 volts direct current (Vdc) (resistive); and 3 amperes at 100 Vac, 28 Vdc (inductive).

3.4.7.3 Bleed valve. Filter housings shall be equipped with a bleed valve to vent pressure and air before and after element change. The bleed plug thread size shall be 0.438-20UNF.

3.4.8 Drain plug. Filter housings shall be equipped with a drain plug. The positioning of the plug shall permit maximum drainage of oil from the housing.

3.4.9 Mounting. The configuration shall include provisions for solid mounting of the filter housings. The mounting provisions shall meet the specified shock requirements (see 3.5.2).

3.4.9.1 Element removal clearance. The filter element shall be removable, without damage, from the filter assembly that has an element removal clearance equal to or less than the value specified (see 3.1).

3.4.10 Welding and allied processes. Welding and allied processes shall be in accordance with MIL-STD-278 for the housing pressure vessel or structural weld, as applicable.

3.5 Performance. Filter housings, bypass relief valves, and differential pressure indicators shall perform as specified when subjected to the examination and tests specified (see 4.3 and 4.4).

3.5.1 Proof pressure. Filter housings shall withstand a proof pressure of 1-1/2 times the operating pressure without evidence of permanent deformation, malfunction, or external leakage.

3.5.2 Shock. Filter housings, bypass relief valves, and differential pressure indicators shall meet the shock requirements of MIL-S-901 for grade A, class 1 equipment.

3.5.3 Housing pressure drop. The pressure drop through the filter housing at maximum rated flow, with dummy element installed, shall not exceed the value specified (see 3.1).

3.5.4 Fatigue impulse. Filter housings, bypass relief valves, and differential pressure indicators shall withstand, without failure, 10 million impulse pressure cycles in accordance with NFP(A) T2.6.1.

3.6 Interchangeability. Parts having the same specification part number shall be interchangeable and shall conform to the dimensional requirements specified for these parts. Parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:20Z
Check the source to verify that this is the current version before use.

3.7 Identification.

3.7.1 Marking. Each filter housing shall be clearly and permanently identified by an attached identification plate containing the following information (except for the ANSI fluid symbol, which may be put on a separate plate alone):

(a) Number and date of this specification.
(b) Filter housing specification part number (see 3.1).
(c) Manufacturer's part number.
(d) Manufacturer's name or trademark.
(e) Manufacturer's CAGE (Commercial and Government Entity) code.
(f) ANSI fluid symbol (see ANSI Y32.10).
(g) Relief valve setting.

3.7.1.1 Inlet and outlet ports. Filter inlet and outlet ports shall be clearly and permanently marked "in" and "out", respectively.

3.7.1.2 Drain and vent ports. Drain and vent ports shall be clearly and permanently marked "drain" and "vent", respectively.

3.7.2 Special marking. Each housing shall incorporate appropriate caution or warning plates, as necessary, to assure lubrication and assembly procedures that will prevent galling of threads during element change.

3.8 Special tools. Special tools shall not be required to install or service the housings. Special tools are defined as those not listed in the Federal Supply Catalog (copies of this catalog may be consulted in the office of the Defense Contract Administration Services Management Area (DCASMA)).

3.9 Workmanship. The housings shall conform to this specification and be free of burrs, sharp edges or defects which may affect their suitability for the service intended. Where dimensions and tolerances may affect the interchangeability or performance of the filter, they shall be held or limited accordingly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:20Z
Check the source to verify that this is the current version before use.

acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Qualification inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Qualification inspection. Qualification inspection shall be conducted at a laboratory acceptable to the qualifying activity. Qualification inspection shall consist of the examination and tests specified in table I. For each housing, tests shall be performed in the sequence shown in table I. Failure of any test shall be cause for rejection.

TABLE I. Qualification inspection.

Examination and tests	Requirements	Test
Filter housing		
Examination	3.4.1, 3.4.2, 3.4.3, 3.4.6, 3.4.7 thru 3.4.9, 3.6, 3.7.1 thru 3.7.1.2, 3.7.2, 3.9	4.6
Proof pressure	3.5.1	4.7.1.1
Shock	3.5.2	4.7.1.2
Housing pressure drop	3.5.3	4.7.1.3
Fatigue impulse	3.5.4	4.7.1.4
Relief valve operation	3.4.7	4.7.1.5
Element removal	3.4.9.1	4.7.1.6
Bypass relief valve and differential pressure indicator		
Examination	3.4.7	4.6
Shock	3.5.2	4.7.1.2
Fatigue impulse	3.5.4	4.7.1.4
Operation	3.4.7	4.7.1.5

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:20Z
Check the source to verify that this is the current version before use.

4.3.1 Qualification by similarity. Qualification on the basis of similarity will be considered by NAVSEA (see 4.3.1.1 and 4.3.1.2).

4.3.1.1 Housings. Qualification approval without additional testing will be considered based on the qualification of a similar other size housing where the only change is the length of filter element housing.

4.3.1.2 Pressure indicators. Qualification approval for a housing with different forms of pressure indicators may be extended from the original housing qualification provided the different pressure indicators have passed qualification tests. The rated fatigue pressure of the indicator must be at least equivalent to the fatigue rating of the housing.

4.3.2 Data submittal.

4.3.2.1 Drawings. Prior to authorization for qualification testing, two sets of assembly drawings shall be furnished with each new size of filter housing submitted for qualification tests. Assembly drawings shall show a cutaway section of details in their normal assembly positions and shall carry part numbers of details and subassemblies. The following data shall be furnished on the assembly drawings:

(a) Outline dimensions of the complete assembly.
(b) Dimensional location of ports, port sizes, and port identification markings.
(c) Dimensional location and size of mounting holes.
(d) Cross-sectional views showing internal flow paths.
(e) Bill of material, listing specifications, grade, and condition, or other data needed to identify the materials proposed.
(f) Complete dismantling procedure and description of tools needed.
(g) Any special installation or operating instruction considered necessary.

Dimensions and tolerances shall be in accordance with ANSI Y14.5.

4.3.2.2 Photographs. A photograph showing the actual impulse curve of the fatigue impulse test (see 4.7.1.4), as indicated by oscilloscope (or equivalent method), shall be included in the qualification test report.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examinations and tests specified in table II, conducted in the order listed. Failure of any test shall be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:20Z
Check the source to verify that this is the current version before use.

TABLE II. Quality conformance inspection.

Examinations and tests	Requirements	Test
Filter housing		
Examination	3.4.1, 3.4.2, 3.4.3, 3.4.7, thru 3.4.9, 3.6, 3.7.1 thru 3.7.1.2, 3.7.2, 3.9	4.6
Proof pressure	3.5.1	4.7.1.1
Relief valve operation	3.4.7	4.7.1.5
Bypass relief valve and differential pressure indicator		
Examination	3.4.7	4.6
Operation	3.4.7	4.7.1.5

4.5 Test conditions.

4.5.1 Test fluid. The hydraulic fluid used for all tests shall be in accordance with MIL-H-17672, grade 2135 TH.

4.5.2 Test fluid temperature. Unless otherwise specified, the test fluid temperature shall be that temperature at which the actual fluid viscosity is 68 centistokes. The temperature shall be maintained to plus or minus 2 degrees Celsius (°C). Viscosity data used to determine the test temperature (approximately 40°C) shall be included in the qualification test report.

4.6 Examination. The filter housing, bypass relief valve, and differential pressure indicator shall be examined to determine compliance with the requirements specified herein with respect to construction, dimensions, visible defects, marking, workmanship, and the presence of a bypass valve indicator, vent and drain port.

4.7 Test methods.

4.7.1 Filter housings. Unless otherwise specified, all tests shall be conducted with filter elements installed.

4.7.1.1 Proof pressure. A proof pressure of 1-1/2 times the operating pressure shall be applied to the filter housing and held for 2 minutes (see 3.5.1). For qualification, this test shall be performed after completion of the shock and impulse tests (see table I). Any evidence of permanent deformation, malfunction, or external leakage shall be cause for failure.

Source: https://assist.dla.mil Downloaded: 2026-10-20T01:20Z
Check the source to verify that this is the current version before use.

4.7.1.2 Shock. Filter housing, bypass relief valve, and differential pressure indicator shall be subjected to a class A shock test in accordance with MIL-S-901. Any evidence of damage, malfunction, or external leakage at the conclusion of the test or during subsequent tests (see 4.7.1.3 through 4.7.1.5) shall be cause for failure.

4.7.1.3 Housing pressure drop. This test shall be conducted with a dummy element installed. Using the test fluid, the maximum pressure drop between the inlet and outlet of the filter housing at rated flow shall not exceed the value specified (see 3.1). Exceeding the values specified shall be cause for failure.

4.7.1.4 Fatigue impulse test. The filter housings, bypass relief valves, and differential pressure indicators which have passed the shock test shall be impulse cycled in accordance with NFP(A) T2.6.1. The cycle test pressure to be used shall be determined by multiplying the rated fatigue pressure by applicable test duration and variability factors specified in NFP(A) T2.6.1 to obtain a 90 percent assurance level based on a 90 percent confidence level. Seals may be replaced during the test. The replacing of seals during impulse testing shall not be considered as a test failure. Any evidence of damage or malfunction shall be cause for failure.

4.7.1.5 Bypass-relief valve and differential pressure indicator operation.

4.7.1.5.1 Bypass valve - qualification. The filter housings shall be tested for bypass valve operation. Prior to this test, a means of blocking flow through the filter elements shall be installed. The pointer of pressure indicator shall be in the green area of the tricolor scale. Pressure shall be applied to the inlet port of the filter at 75 percent of cracking pressure by means of a power-driven pump and shall be increased in increments of 5 lb/in^2 or less until the cracking pressure of the bypass valve is reached (see 3.1 and 3.4.7.1). The pressure indicator pointer movement on tricolor scale shall be observed and recorded during the pressure application. The pressure shall be held constant at each increment and leakage rate noted for at least 1 minute. The pressure at which the leakage rate through the bypass valve is 10 milliliters (mL) per minute shall be the valve cracking pressure. Note the position of the pressure indicator pointer at the cracking pressure. The pressure shall be increased until rated flow is obtained through the bypass valve. Rated flow shall occur at a differential pressure not exceeding the value specified (see 3.1). Note the position of pressure indicator on the tricolor scale. The pressure shall then be reduced in 5 lb/in^2 or smaller increments until leakage through the valve does not exceed 8 mL per minute. The resulting pressure shall be considered the bypass valve reseat pressure. The pressure indicator pointer shall be observed during the depressurization of filter housing. Non-conformance to 3.4.7.2 shall constitute failure of this test.

4.7.1.5.2 Bypass valve - quality conformance. The filter housings shall be tested for conformance to leakage requirements at cracking pressure. Leakage in excess of these requirements shall be cause for failure.

4.7.1.5.3 Pressure indicator - quality conformance. Operation of pressure indicator shall be tested for conformance to visual requirements of 3.4.7.2.1 and 3.4.7.2.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:20Z
Check the source to verify that this is the current version before use.

4.7.1.6 Element removal. With the filter housing secured so that element removal clearance is restricted to the value specified (see 3.1), remove the filter element from the housing. Any evidence of damage, or inability to remove the element shall be cause for failure of the test.

4.8 Inspection of packing. Sample packs, and the inspection of the preservation packing, and marking for shipment, stowage, and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.5.)

5.1 General

5.1.1 Navy shipboard stowage fire-retardant requirements.

(a) Treated lumber and plywood. Unless otherwise specified (see 6.2), all lumber and plywood including laminated veneer material used in shipping containers and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Levels A and B - Type II - weather resistant.
Category 1 - general use.
Level C - Type I - non-weather resistant.
Category 1 - general use.

(b) Fiberboard. Fiberboard used in the construction of class-domestic, non-weather resistant fiberboard, cleated fiberboard boxes including interior packaging forms shall meet the flame spread index and the specific optic density requirements of PPP-F-320.

5.2 Preservation. Preservation shall be level A, C, or commercial, as specified (see 6.2).

5.2.1 Level A. Filter housing and differential pressure indicator, internal surfaces and components in contact with hydraulic fluid shall be cleaned to a grade A/B level ultrasonically (or by an equivalent method). Grade A/B cleanliness is defined as that degree of cleanliness which results in free flow paths and internal surfaces free of grease and oil (except approved preservatives and assembly lubricants), flux, loose particles visible to the unaided eye, oxides and tap water residues having visibly distinguishable particles, temper and oxide films resulting from heating operations which are of visible thickness, metal chips, grit, dirt, and other foreign materials which could be detrimental to the operation of hydraulic equipment. This level of cleanliness for particulate matter shall be determined by visual examination. The cleaning method utilized shall not have any deleterious effects on any material (for example, metals,plastics and elastomers). Immediately after cleaning, the units shall be dried, assembled as necessary, and unless otherwise specified (see 6.2), flushed with preservative oil conforming to MIL-H-6083, drained, and tagged as specified in 5.4.2. Openings shall be sealed with closures conforming to MIL-C-5501.

Source: https://assist.dla.mil - Downloaded: 2026-10-20T01:20Z
Check the source to verify that this is the current version before use.

5.2.1.1 Unit pack. Unit packs (unit containers) shall be in accordance with MIL-STD-2073-1, appendix F, table I. Unless otherwise specified (see 6.2), container selection shall be at the contractor's option. Containers shall be of type weather resistant and conform to closure method V in accordance with the appendix of the box specification.

5.2.2 Level C. Filters shall be preserved as specified for level A except that the unit containers specified in 5.2.1.1 shall be of the non-weather resistant type, class, or variety as applicable. Fiberboard box closure shall be in accordance with method I of the box specification, using pressure sensitive tape.

5.2.3 Commercial. Commercial preservation shall be in accordance with ASTM D 3951 except that filter housings shall be cleaned as specified in 5.2.1.

5.3 Packing. Packing shall be level A, B, C, or commercial as specified (see 6.2).

5.3.1 General. Shipping containers shall contain identical quantities of identical material and shall be of minimum weight and cube, similar construction, and of uniform size.

5.3.2 Level A, B, and C containers. Materials preserved as specified in 5.2 shall be packed in shipping containers for the level of packing specified (see 6.2), in accordance with MIL-STD-2073-1, appendix C, table VII. Unless otherwise specified (see 6.2), container selection shall be at the contractor's option.

5.3.2.1 Closure, gross weight, and waterproofing.

5.3.2.1.1 Closure. Container closure, reinforcing, or banding shall be in accordance with the applicable container specification or appendix thereto except that class weather resistant fiberboard boxes shall be closed in accordance with method V and reinforced with nonmetallic or tape banding, and non-weather resistant fiberboard boxes shall be closed in accordance with method I using pressure sensitive tape.

5.3.2.1.2 Weight. Wood, plywood, and cleated type containers exceeding 200 pounds gross weight shall be modified by the addition of skids in accordance with MIL-STD-2073-1 and the applicable container specification or appendix thereto.

5.3.2.1.3 Waterproofing. Unless otherwise specified (see 6.2), level A and, when specified (see 6.2), level B shipping containers shall be provided with caseliners, linings, wraps, or shrouds in accordance with the waterproofing requirements of MIL-STD-1186.

5.3.3 Commercial. Material preserved as specified (see 6.2) shall be packed for shipment in accordance with ASTM D 3951 and herein.

5.3.3.1 Container modification. Shipping containers exceeding 200 pounds gross weight shall be provided with the minimum of 3- by 4-inch nominal wood skids laid flat, or a skid- or sill-type base that will support the material and facilitate handling by mechanical handling equipment during shipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:20Z
Check the source to verify that this is the current version before use.

5.4 Marking. Marking shall be as specified in 5.4.1.

5.4.1 Level A, B, C, and commercial. In addition to any special marking required (see 6.2), level A, B, and C interior packs and shipping containers shall be marked in accordance with MIL-STD-2073-1, appendix F, and commercial interior packs and shipping containers shall be marked in accordance with ASTM D 3951. In addition, bar coding shall be applied in accordance with MIL-STD-2073-1.

5.4.2 Special marking. Preserved filter housings shall be tagged, the individual unit container and exterior shipping container marked or labeled with the following:

(a) Tags: "Do not disassemble. The interior of this item has been cleaned and examined to grade A/B cleanliness requirements of MIL-F-24724, and is coated with corrosion preventative compound (brand name and type of Military specification number), manufactured by (name of company). It was installed on (date) and should be replaced 1 year from date if still in storage. Preservation replacement does not require disassembly of the unit. Represervation can be done by (state procedure)."

When the preservative fluid conforms to MIL-H-6083 as specified in 5.2.1, the following additional information shall be added to the tag: "Removal of residual preservative fluid prior to installation is not required."

(b) Labels of marking: "The item contained herein has been cleaned and examined to grade A/B cleanliness requirements of MIL-F-24724 and is coated with corrosion preventative compound (brand name and type or Military specification number), manufactured by (name of company). It was installed on (date) and should be replaced 1 year from date if still in storage. Preservation replacement does not require disassembly of the unit. Represervation can be done by (state procedure)."

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The filter housings covered by this specification are intended for use in ship petroleum oil, water glycol, and phosphate ester fluid hydraulic systems. For application to systems using other than these fluids, users should ensure compatibility of the fluid used with the filter housings and elements.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(c) Title, number, and date of applicable specification sheet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:20Z
Check the source to verify that this is the current version before use.

(d) Filter housing specification part number (see 1.2).
(e) Bypass relief valve setting (see 3.4.7.1).
(f) Pressure indicator type (see 3.4.7.2).
(g) Fire-retardant material requirements (see 5.1.1).
(h) Level of preservation, packing, and marking required (see 5.2 through 5.4).
(i) Flushing of filter housings with preservative oil (see 5.2.1).
(j) Container selection (see 5.3.2).
(k) Waterproofing of shipping containers (see 5.3.2.1.3).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List No. 24724 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 and information pertaining to qualification of products may be obtained from that activity. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3.1).

6.3.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

6.4 Definitions. The term "filter housing" used throughout this specification is a filter housing with differential pressure indicator, vent and drain, but without filter elements.

6.5 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.6 Subject term (key word) listing.

Bleed valve
Bypass valve
Full-flow
Hydraulic fluids
Seals

Preparing activity:
Navy - SH
(Project 4330-N051)

☆U.S. GOVERNMENT PRINTING OFFICE: 1990 - 704-034/30376
Source: https://assist.dla.mil -- Downloaded: 2020-10-26 10:20Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

COMMANDER
NAVAL SEA SYSTEMS COMMAND (SEA 5523)
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20362-5101

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

COMMANDER
NAVAL SEA SYSTEMS COMMAND (SEA 5523)
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20362-5101

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:20Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-24724(SH)	FILTER HOUSINGS (HYDRAULIC) AND DIFFERENTIAL PRESSURE INDICATORS, GENERAL SPECIFICATION FOR

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 1426 82 MAR PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil/online/2245.pdf 2020-10-20T01:20Z
Check the source to verify that this is the current version before use.